SEMINAR REPORT

A seminar Report on

AI Based Rainwater Harvesting Potential Predector

Submitted in partial fulfilment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960288	388588	Hemangi Shivaji Suryawanshi
2	24610960289	388589	Kalyani Naresh Thakare
3	24610960293	388593	Vidhi Shekhar Vibhandik
4	25610960008	388606	Shrawani Nutan Gautam

FOR THE ACADEMIC YEAR: 2026 – 2027

DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code: 0059)

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMENT OF COMPUTER ENGINEERING

CERTIFICATE

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960288	388588	Hemangi Shivaji Suryawanshi
2	24610960289	388589	Kalyani Naresh Thakare
3	24610960293	388593	Vidhi Shekhar Vibhandik
4	25610960008	388606	Shrawani Nutan Gautam

of **CO5K** Semester of Diploma in **Computer Engineering** of **S.S.V.P.S's B. S. Deore Polytechnic, Dhule** (Institute Code: 0059) has completed the term work satisfactorily in Course **Seminar Presentation and Project Initiation (Course Code: 315003)** for the academic year **2026 to 2027** as prescribed in the curriculum.

Place : Dhule

Date : 17/10/2026

Signature:

Name : Mr.Nishad D. Patel

Subject Teacher

Signature:

Name : Mr.Nishad D. Patel

Head of the Department

Seal of Institution

TABLE OF CONTENT

Sr. No.	Chapter	Page No.
	Abstract	i
	List of Figures	ii
	List of Tables	iii
1	Introduction	1
2	Literature Review	2 - 5
3	Information/Chapters	6 - 9
4	Advantages and Disadvantages	10
5	Conclusion	11
6	Project Initiation:	12
	6.1) Description of problem statement.	12
	6.2) Scope and objectives.	12
	6.3) Stake holder	13
	6.4) Platform/ Equipment/ Resources identification.	13-14
7	Seminar Presentation	15 - 25
	Bibliography	26
	References	27

ABSTRACT

Rainwater is one of the most valuable natural resources required for human life, agriculture, industries, and environmental sustainability. Due to increasing population, urbanization, irregular rainfall, and growing water demand, water scarcity has become a major concern in many regions. Rainwater harvesting is an effective method of collecting and storing rainwater for future use. However, estimating the amount of rainwater that can be collected from a particular area is important for planning an efficient rainwater harvesting system.

This seminar report presents the concept of AI Based Rainwater Harvesting Potential Predictor, a system that uses Artificial Intelligence (AI) and Machine Learning (ML) techniques to estimate the potential amount of rainwater that can be harvested from a given area. The system considers important parameters such as rainfall data, rooftop area, roof surface type, runoff coefficient, and seasonal weather conditions to predict the quantity of rainwater that can be collected.

The report explains the working principle, system architecture, techniques, technologies, advantages, and limitations of an AI-based rainwater harvesting prediction system. Machine learning algorithms can analyse historical rainfall data and identify patterns to support more accurate predictions. The system can help households, farmers, educational institutions, and urban planners make better decisions regarding rainwater collection and storage.

The report also includes a literature review of related research, a project initiation plan, the problem statement, scope, objectives, stakeholders, and required resources. Overall, an AI Based Rainwater Harvesting Potential Predictor can support efficient water resource management, reduce dependence on external water sources, and encourage sustainable water conservation practices.

LIST OF FIGURES

Fig. No.	Title	Page No.
3.4	Working of AI Based Rainwater Harvesting Potential Predictor	7
3.5	Architecture of AI Based Rainwater Harvesting Potential Predictor	8

LIST OF TABLES

Table No.	Title	Page No.
2.1	Literature Review	5
6.1	Platform, Equipment and Resources	14

CHAPTER 1
INTRODUCTION

Artificial Intelligence (AI) is a rapidly growing field of computer science that enables computer systems to analyse data, identify patterns, and make predictions. Machine Learning (ML), a branch of AI, helps computers learn from historical data and predict future outcomes. These technologies are useful in weather forecasting, agriculture, and environmental management.

Water is one of the most essential natural resources required for human life, agriculture, industries, and environmental sustainability. Due to increasing population, urbanization, and irregular rainfall, water scarcity has become a major concern in many regions. Therefore, effective water conservation and management are necessary.

Rainwater harvesting is a method of collecting and storing rainwater from rooftops and other surfaces for future use. The amount of water collected depends on factors such as rainfall, rooftop area, roof surface type, and runoff coefficient. Estimating this quantity helps in planning suitable rainwater collection and storage systems.

The AI Based Rainwater Harvesting Potential Predictor is a proposed system that uses Artificial Intelligence and Machine Learning techniques to estimate the potential amount of rainwater that can be harvested from a particular area. It analyses rainfall data, rooftop area, and other relevant parameters to predict the approximate quantity of collectible rainwater.

Traditional methods mainly depend on manual calculations using rainfall and rooftop measurements. An AI-based approach can analyse historical data, identify patterns, and support better prediction and planning. This system can help households, farmers, and institutions understand their rainwater harvesting potential.

Overall, the proposed system aims to promote efficient water conservation, reduce water wastage, and support sustainable water resource management.

CHAPTER 2
LITERATURE REVIEW

- **Review of Research Paper**

1. **Rainwater Harvesting Potential Zone Identification Using GIS – Machine Learning Approach and Comparison of Their Performance (2026)**

   This study develops a GIS and Machine Learning framework to identify suitable rainwater harvesting locations in the Choke Watershed, Ethiopia. Random Forest, Gradient Boosting, and XGBoost algorithms were compared using environmental and hydrological factors. The study found that Gradient Boosting and XGBoost achieved an AUC value of 0.97. It demonstrates the usefulness of machine learning for identifying suitable rainwater harvesting areas.

2. **Integrating Machine Learning and Geo-Hydrological Modeling to Identify Rainwater Harvesting Potential Zones in the Upper Tekeze Basin (2026)**

   This research combines hydrological modelling, Random Forest, and geospatial analysis to identify suitable rainwater harvesting zones in Ethiopia. The study considers rainfall, slope, drainage density, and other environmental factors. The results show that combining machine learning with hydrological modelling can improve the identification of suitable harvesting locations.

3. **Integrated Machine Learning and AHP-Based Framework for Spatial Delineation of Rainwater Harvesting Potential (2026)**

   This study uses GIS, the Analytical Hierarchy Process (AHP), and machine learning algorithms to identify suitable rainwater harvesting areas in Jaipur district, Rajasthan. It considers rainfall, elevation, slope, soil, land use, and drainage characteristics. The research highlights the importance of combining multiple environmental parameters when assessing rainwater harvesting potential.

4. **Assessing the Performance of Machine Learning and AHP Models for Rainwater Harvesting Potential Zone Identification in Hilly Region, Bangladesh (2025)**

   This research compares Random Forest, Boosted Regression Trees, K-Nearest Neighbors, Naive Bayes, and AHP for identifying suitable rainwater harvesting zones. The study considers eleven parameters, including rainfall, slope, elevation, geology, and drainage density. Boosted Regression Trees and Random Forest achieved AUC values of 0.93, demonstrating the potential of machine learning for site suitability prediction.

5. **Estimation of Roof Top Rain Water Harvesting Potential: A Proposed Solution to Water Scarcity in Shrigonda Town, Maharashtra, India (2024)**

   This study estimates rooftop rainwater harvesting potential in Shrigonda town, Maharashtra, using rooftop area and rainfall data. It applies a conventional rainwater estimation formula to calculate the amount of water that could be collected. The findings highlight the importance of rooftop measurements and rainfall information for planning rainwater harvesting systems in Maharashtra.

6. **Addressing Water Scarcity Challenges Through Rainwater Harvesting: A Case Study on Purulia, India (2024)**

   This research uses Artificial Neural Networks (ANN), Random Forest, and geospatial techniques to identify rainwater harvesting potential zones in Purulia, India. Important factors include rainfall, soil, slope, land cover, runoff potential, and drainage density. The study reports that Random Forest achieved nearly 99% accuracy in its suitability classification, showing the potential of machine learning in water resource planning.

7. **Urban Rooftop Area Assessment for Estimation of Rooftop Rainwater-Harvesting Potential (2020)**

   This study by T. Mouli, K. Krupavathi, and Ch. S. Ganeswara Rao uses satellite imagery and GIS to estimate rooftop areas for rainwater harvesting. The researchers validated their measurements using manual surveys of 112 household systems. The study shows that rooftop area estimation and rainfall analysis are important for calculating rainwater harvesting potential.

8. **Feasibility of Roof Top Rainwater Harvesting Potential – A Case Study of South Indian University (2021)**

   This research examines rooftop rainwater harvesting at a university in South India. It analyses rainfall, rooftop runoff, water quality, and storage tank requirements using field observations and GIS techniques. The study estimates that approximately 113,678.9 cubic metres of rainwater could be collected annually. It highlights the importance of rainfall analysis and storage planning.

9. **Estimation of Groundwater Recharge Potential Using Rooftop Rainwater Harvesting: Case Study from Pune Urban Area, India (2022)**

   This study by N. Vaddadi, C. Vansarochana, and V. Raghavan investigates rooftop rainwater harvesting as a method of groundwater recharge in Pune. The researchers estimated annual rooftop runoff and examined its potential contribution to groundwater resources. The study indicates that rooftop rainwater harvesting can support groundwater replenishment and improve urban water management.

10. **Identifying Potential Zones for Rainwater Harvesting Interventions for Sustainable Intensification in the Semi-Arid Tropics (2022)**

    This research examines rainfall-runoff relationships and identifies areas suitable for rainwater harvesting in the Indian semi-arid tropics. It analyses rainfall, surface runoff, and the effects of harvesting interventions on groundwater availability and soil loss. The findings highlight the importance of rainfall and runoff data when selecting suitable water conservation strategies.

11. **Assessing Potential for Rooftop Rainwater Harvesting: An Option for Sustainable Rural Domestic Water Supply in Arid Region of Haryana (2017)**

    This study by Omvir Singh and Sushila Turkiya estimates rooftop rainwater harvesting potential in a village in Haryana. The researchers surveyed 763 households and considered rooftop areas, rainfall, and runoff coefficients for different building surfaces. The research demonstrates how household-level data can help estimate rainwater availability and support rural water conservation planning.

12. **Planning of Rooftop Rainwater Harvesting Structure Based on GIS Approach (2018)**

    This study by P. A. Pandya and colleagues uses Google Earth and Quantum GIS to estimate rooftop rainwater harvesting potential at Junagadh Agricultural University, Gujarat. It evaluates rooftop areas, rainfall, water demand, and storage requirements. The study demonstrates that GIS-based assessment can help determine the quantity of collectible rainwater and support the planning of suitable storage structures.

Table No 2.1: Literature review:

Sr. No.	Paper (Author, Year)	Approach Type	Technologies Used	Key Objective
1	Choke Watershed Study (2026)	Choke Watershed Study (2026)	Choke Watershed Study (2026)	Choke Watershed Study (2026)
2	Upper Tekeze Basin Study (2026)	Upper Tekeze Basin Study (2026)	Upper Tekeze Basin Study (2026)	Upper Tekeze Basin Study (2026)
3	Jaipur District Study (2026)	Jaipur District Study (2026)	Jaipur District Study (2026)	Jaipur District Study (2026)
4	Bangladesh Hilly Region Study (2025)	Bangladesh Hilly Region Study (2025)	Bangladesh Hilly Region Study (2025)	Bangladesh Hilly Region Study (2025)
5	Shrigonda Study (2024)	Shrigonda Study (2024)	Shrigonda Study (2024)	Shrigonda Study (2024)
6	Purulia Study (2024)	Purulia Study (2024)	Purulia Study (2024)	Purulia Study (2024)
7	Mouli et al. (2020)	Mouli et al. (2020)	Mouli et al. (2020)	Mouli et al. (2020)
8	South Indian University Study (2021)	South Indian University Study (2021)	South Indian University Study (2021)	South Indian University Study (2021)
9	Vaddadi et al. (2022)	Vaddadi et al. (2022)	Vaddadi et al. (2022)	Vaddadi et al. (2022)
10	Semi-Arid Tropics Study (2022)	Semi-Arid Tropics Study (2022)	Semi-Arid Tropics Study (2022)	Semi-Arid Tropics Study (2022)
11	Singh and Turkiya (2017)	Singh and Turkiya (2017)	Singh and Turkiya (2017)	Singh and Turkiya (2017)
12	Pandya et al. (2018)	Pandya et al. (2018)	Pandya et al. (2018)	Pandya et al. (2018)

CHAPTER 3
INFORMATION ON SEMINAR

3.1 Overview of AI Based Rainwater Harvesting Potential Predictor

AI Based Rainwater Harvesting Potential Predictor is a system that estimates the amount of rainwater that can be collected and stored from rooftops or other suitable surfaces. The system analyses important parameters such as rainfall, rooftop area, roof surface type, and runoff coefficient to calculate the potential quantity of harvestable rainwater. It uses Artificial Intelligence (AI) and Machine Learning (ML) techniques to analyse historical rainfall data and identify patterns for prediction.

This system is useful for households, farmers, educational institutions, and urban planners. It helps estimate rainwater availability, plan suitable storage systems, and improve water resource management. By predicting rainwater harvesting potential, the system can support water conservation, reduce water wastage, and encourage sustainable use of water resources.

3.2 AI and Machine Learning in Rainwater Harvesting Prediction

Artificial Intelligence (AI) enables computer systems to analyse data and make predictions. Machine Learning (ML), a branch of AI, learns patterns from historical rainfall records and environmental data. In rainwater harvesting prediction, these techniques can help estimate the amount of rainwater that can be collected from a particular area.

The system considers parameters such as rainfall, rooftop area, roof surface type, and runoff coefficient. Machine learning models analyse these inputs to predict the approximate quantity of harvestable rainwater. The accuracy of the prediction depends on the quality of the data and the model used.

3.3 Need for AI Based Rainwater Harvesting Potential Prediction

Rainwater harvesting is an important method of conserving water, especially in areas facing water scarcity and irregular rainfall. Estimating the amount of rainwater that can be collected helps people plan suitable harvesting systems and storage tanks. An AI-based prediction system can support households, farmers, educational institutions, and urban planners in making better water management decisions. It can reduce dependence on manual estimation, improve planning, and encourage efficient use of rainwater. Therefore, this system can contribute to sustainable water conservation and management.

3.4 Working Principle:

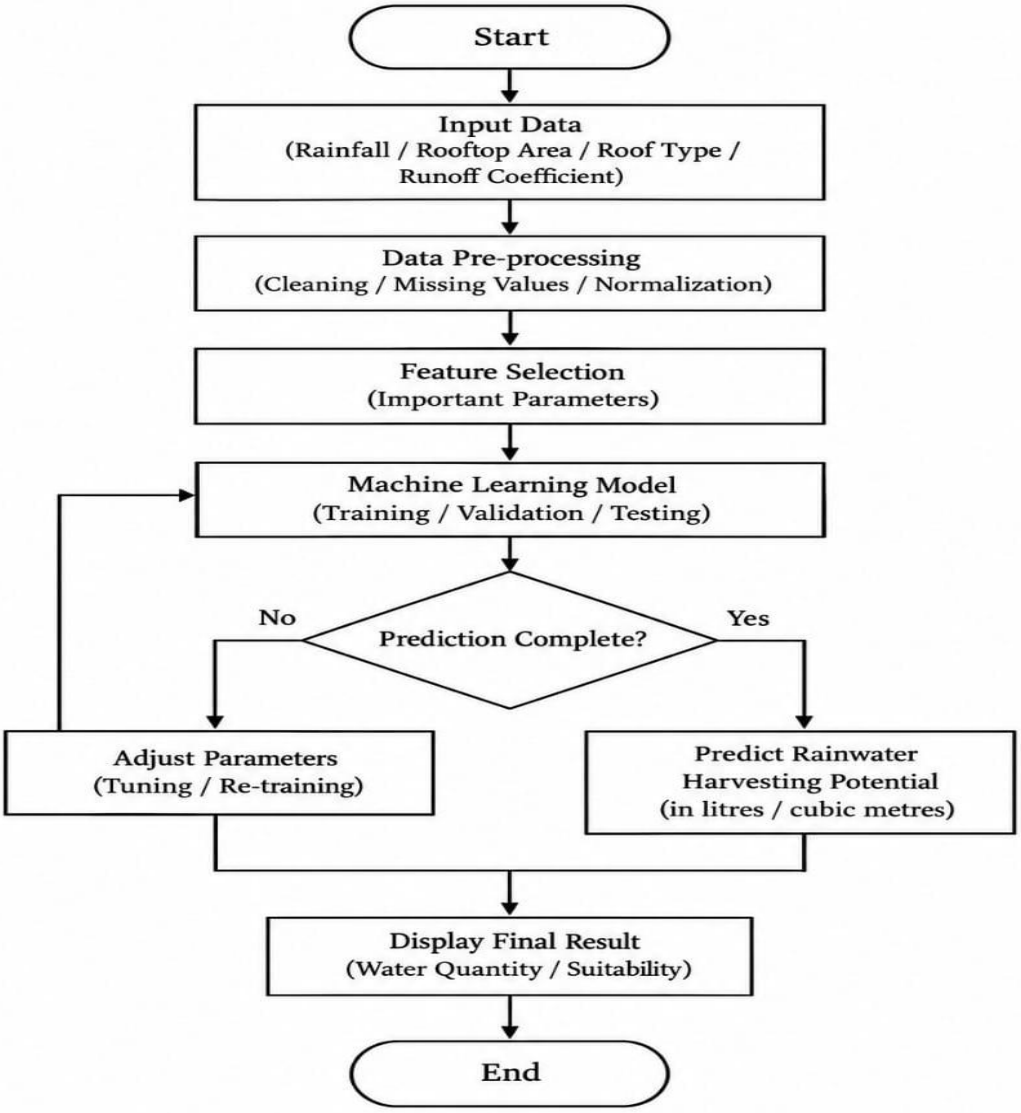


Fig. 3.4: Working of AI Based Rainwater Harvesting Potential Predictor[2]

The working of the AI Based Rainwater Harvesting Potential Predictor consists of the following steps:

1. **Start:** The process begins when the user wants to estimate rainwater harvesting potential.
2. **Input Data:** The user provides rainfall data, rooftop area, roof type, runoff coefficient, and other relevant parameters.
3. **Data Pre-processing:** The collected data is cleaned and organized by handling missing values and preparing it for analysis.
4. **Feature Selection**: Important parameters such as rainfall, rooftop area, and runoff coefficient are selected for prediction.

5. **Machine Learning Model**: A trained model, such as Random Forest, Gradient Boosting, or XGBoost, analyses the input data to estimate rainwater harvesting potential.
6. **Prediction:** The model predicts the approximate quantity of rainwater that can be collected from the given rooftop or area.
7. **Result Calculation:** The predicted quantity is calculated and represented in litres or cubic metres.
8. **Display Final Result:** The estimated rainwater harvesting potential is displayed to the user to support water collection and storage planning.
9. **End:** The process ends after the prediction result is displayed.

3.5 System Architecture and Main Components:

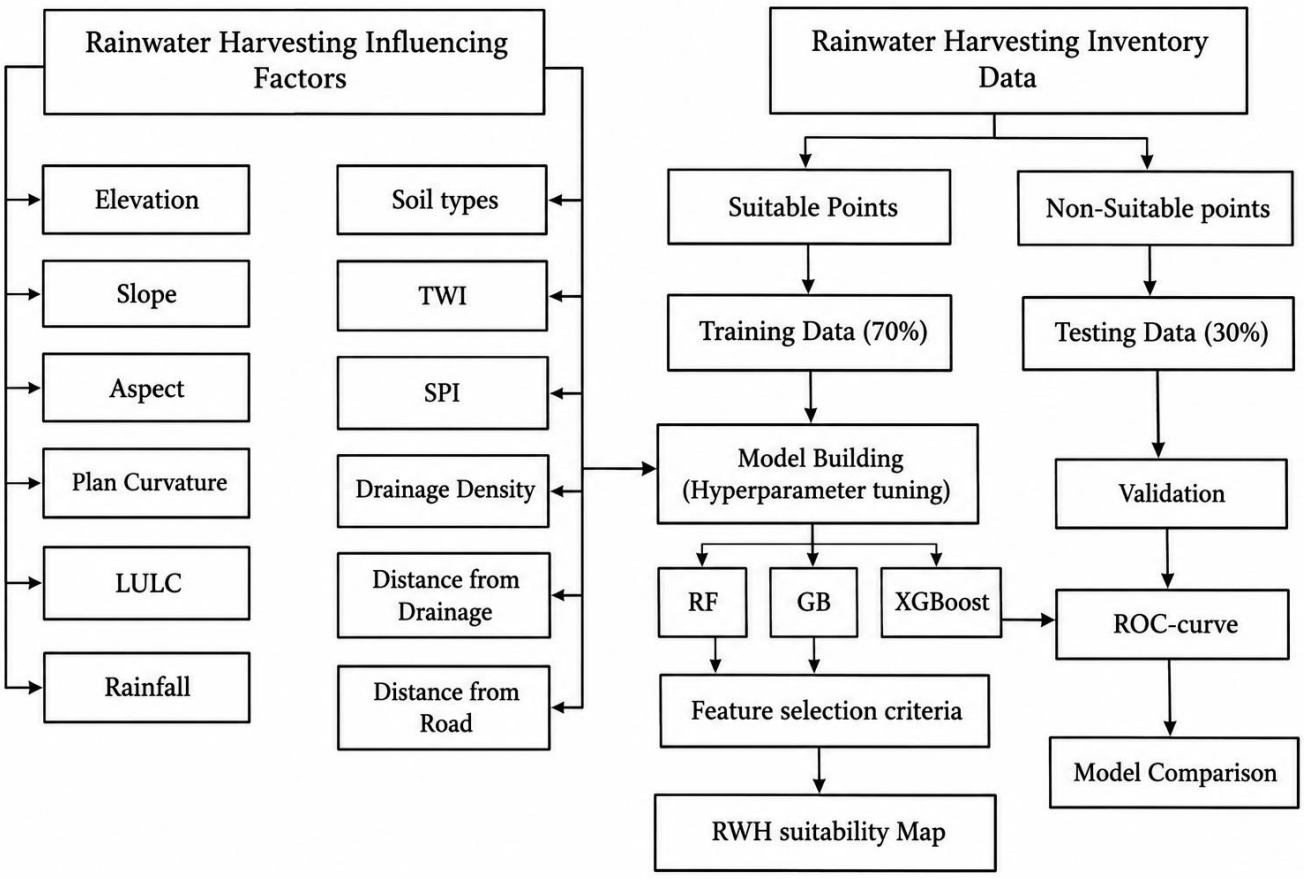


Fig. 3.5: Architecture of AI-Based Rainwater Harvesting Potential Predictor[1]

The system architecture shows how different components of the AI-Based Rainwater Harvesting Potential Predictor work together. It processes rainfall, rooftop area, and other input data using an AI/ML model. Finally, it displays the estimated quantity of harvestable rainwater in litres.

3.6 Techniques and Technologies Used:

- **Machine Learning:** Algorithms such as Random Forest, Gradient Boosting, and XGBoost can analyse rainfall and environmental data to predict rainwater harvesting potential.
- **GIS Technology:** Geographic Information System (GIS) helps identify suitable areas for rainwater harvesting using geographical and environmental information.
- **Rainfall Data Analysis:** Historical rainfall records are analysed to estimate the amount of rainwater available for collection.
- **Data Pre-processing:** Missing values are handled and input data is cleaned and organized before prediction.

3.7 Performance Evaluation Metrics:

The performance of the prediction system can be evaluated using the following metrics:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual rainwater quantities.
- **Root Mean Square Error (RMSE):** Measures prediction errors and gives greater weight to larger errors.
- **R^2 Score:** Indicates how well the model explains variations in the actual rainwater harvesting potential.
- **Accuracy of Estimation**: Shows how closely the predicted rainwater quantity matches the actual or calculated quantity.

3.8 Example of Detection:

Suppose a rooftop has an area of 100 m^2, annual rainfall of 800 mm (0.8 m), and a runoff coefficient of 0.8.

- **Formula:**

  Rainwater Harvesting Potential = Rainfall (m) × Roof Area (m^2) × Runoff Coefficient
- **Calculation:** = $0.8 \times 100 \times 0.8 = 64\ m^3$
- **Result:** The estimated rainwater harvesting potential is 64,000 litres per year.

3.9 Challenges in Detection:

- Irregular Rainfall: Rainfall varies by season and location.
- Data Availability: Incomplete or incorrect data affects prediction.
- Model Accuracy: Different ML models may produce different results.
- Climate Changes: Changing weather conditions affect rainwater availability.
- Storage Planning: Selecting the correct storage capacity can be challenging.

CHAPTER 4
ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Helps to estimate the amount of rainwater that can be collected from rooftops.
2. Supports water conservation and reduces water wastage.
3. Saves time by reducing manual calculations.
4. Analyses large amounts of rainfall data quickly.
5. Machine learning models improve rainwater potential prediction.
6. Helps households and institutions plan suitable water storage systems.

4.2 Disadvantages:

1. Prediction accuracy depends on the quality of input data.
2. Predicted rainwater quantity may differ from actual collection.
3. Developing and maintaining the system may require additional resources.
4. Machine learning models require suitable training data.
5. Unexpected weather changes can affect prediction results.
6. Limited storage capacity may cause water overflow during heavy rainfall.

CHAPTER 5
CONCLUSION

AI Based Rainwater Harvesting Potential Predictor is a useful system for estimating the amount of rainwater that can be collected from rooftops and other suitable surfaces. Water scarcity is becoming a major concern due to increasing population, urbanization, irregular rainfall, and growing water demand. Therefore, rainwater harvesting is an effective method of conserving water and ensuring its availability for future use. By using Artificial Intelligence (AI) and Machine Learning (ML) techniques, the proposed system can analyse rainfall data and other important parameters to estimate rainwater harvesting potential.

The system considers factors such as rainfall, rooftop area, roof surface type, and runoff coefficient to calculate the approximate quantity of harvestable rainwater. Machine learning algorithms can analyse historical data, identify patterns, and support better predictions. The estimated results can help households, farmers, educational institutions, and urban planners select suitable rainwater harvesting methods and plan storage facilities according to their requirements.

The proposed system can reduce the effort required for manual calculations and make rainwater harvesting planning more systematic and efficient. However, the accuracy of predictions depends on the quality and availability of input data, the selection of suitable algorithms, and changing weather conditions. Proper data preprocessing, model training, testing, and regular evaluation are necessary to improve the reliability of the system.

Overall, AI Based Rainwater Harvesting Potential Predictor has the potential to support better water resource management and promote sustainable water conservation practices. It can help people understand how much rainwater may be collected from a particular area and encourage the effective use of available rainfall. In the future, the system can be improved by including real-time weather data, advanced machine learning algorithms, and geographical information to provide more reliable predictions and support efficient rainwater harvesting planning.

CHAPTER 6
PROJECT INITIATION

6.1 Description of Problem Statement:

Water scarcity is becoming a major problem due to increasing population, urbanization, irregular rainfall, and growing water demand. Rainwater harvesting is an effective method of collecting and storing rainwater for future use. However, estimating the amount of rainwater that can be collected from a particular rooftop or surface is important for planning a suitable harvesting system. Traditional methods mainly depend on manual calculations using rainfall, rooftop area, and runoff coefficient. These calculations may require time and accurate input data. Therefore, a system is needed that can use Artificial Intelligence (AI) and Machine Learning (ML) techniques to analyse important parameters and predict the potential amount of rainwater that can be harvested.

6.2 Scope of the Project:

The purpose of this project is to study and propose an AI Based Rainwater Harvesting Potential Predictor that estimates the quantity of rainwater that can be collected from rooftops or other suitable surfaces. The system considers parameters such as rainfall, rooftop area, roof surface type, and runoff coefficient. It can use historical rainfall data and machine learning techniques to analyse the input data and predict the approximate amount of harvestable rainwater. The predicted result can help households, farmers, educational institutions, and urban planners plan suitable rainwater collection and storage systems. The project focuses on improving water conservation and supporting efficient water resource management.

Objectives of the Project:

1. To study existing methods of rainwater harvesting potential estimation.
2. To collect and analyse rainfall data and other relevant parameters.
3. To understand the use of Artificial Intelligence and Machine Learning in rainwater prediction.
4. To estimate the amount of rainwater that can be collected from a given rooftop area.
5. To support the planning of suitable rainwater storage systems.
6. To promote water conservation and sustainable water resource management.

6.3 Stakeholders:

The stakeholders of this project are the people who may use the system, develop it, or benefit from its predictions.

1. **Households:** Users who want to estimate the amount of rainwater that can be collected from their rooftops.
2. **Farmers:** Users who can use rainwater predictions for agricultural water planning.
3. **Educational Institutions:** Institutions that can estimate rainwater collection potential for their buildings.
4. **Urban Planners:** Professionals who can use the estimates for water conservation and planning.
5. **Developers:** Team members responsible for designing, developing, and testing the system.
6. **Project Guide:** The faculty member who guides the project and evaluates its progress.

6.4 Platform / Equipment / Resources:

1. **Platform:**

   The system can be developed on Windows or Linux using the Python programming language. Visual Studio Code can be used for writing and testing the program. Python libraries can be used for data processing, analysis, and machine learning model development.

2. **Equipment:**

- Computer or Laptop: Required for developing and running the application.
- Processor: A standard modern processor suitable for Python programming and data analysis.
- RAM: Minimum 8 GB recommended for development and model training.
- Internet Connection: Required for collecting rainfall datasets and downloading libraries.
- Storage: Sufficient storage space for source code, datasets, and trained model files.
- Software Tools: Python, Visual Studio Code, and suitable machine learning libraries.
- Data Resources: Historical rainfall records, rooftop area measurements, and runoff coefficient values required for prediction.

3. Resources:

Table No. 6.1: Resources[1]

Resources	Name	Purpose
Operating System	Windows 11	To develop and run the application and testing environment.
Programming Language	Python 3.11	Primary language used to implement the application and prediction logic.
Code Editor	Visual Studio Code	Code editing, debugging, and development.
ML Library	scikit-learn	Building and evaluating machine learning models.
Machine Learning Algorithms	Random Forest, Gradient Boosting, XGBoost	Analysing rainfall data and predicting rainwater harvesting potential.
Data Handling	pandas, NumPy	Loading, cleaning, and processing rainfall datasets.
Data Visualization	Matplotlib	Displaying rainfall patterns and prediction results through graphs.
Dataset	Historical rainfall data	Training and testing the prediction model.
Input Data	Rainfall, rooftop area, runoff coefficient	Providing the parameters required to estimate rainwater harvesting potential.
Development Equipment	Computer or Laptop	Developing, running, and testing the system.
Internet Connection	Internet access	Downloading datasets, Python libraries, and required resources.

CHAPTER 7

SEMINAR PRESENTATION

	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2026-2027*

SEMINAR AND PROJECT INITIATION COURSE		
Sem : CO 5 K	**SPI – 315003**	**Date :**
Topic : AI Based Rainwater Harvesting Potential Predector		

Roll No.	Name of Student
101	Hemangi Shivaji Suryawanshi
102	Kalyani Naresh Thakare
106	Vidhi Shekhar Vibhandik
119	Shrawani Nutan Gautam

Contents

- Certificate
- Abstract

1. Introduction
2. Literature Review
3. Information to Seminar topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :
   - 6. 1 Description of problem statement.
   - 6. 2 Scope and objectives.
   - 6. 3 Stake holder
   - 6. 4 seminar exsicussion plan
   - 6.5 Platform/ Equipment/ Resources identification.
1. Bibliography
2. References

Abstract

- Water scarcity and irregular rainfall have increased the need for effective water conservation. Rainwater harvesting helps collect and store rainwater for future use.
- AI-Based Rainwater Harvesting Potential Predictor estimates collectible rainwater using rainfall, rooftop area, and runoff coefficient.
- This report reviews 12 research papers on rainwater harvesting, GIS, and Machine Learning techniques such as Random Forest, Gradient Boosting, and XGBoost.
- It explains the system's working, techniques, advantages, limitations, and project initiation plan to support efficient water conservation.

1. Introduction

- Water is one of the most essential natural resources required for human life, agriculture, and environmental sustainability. Increasing population, urbanization, and irregular rainfall have made water scarcity a major concern.
- Rainwater harvesting is an effective method of collecting and storing rainwater for future use. Traditional methods estimate harvesting potential using rainfall, rooftop area, and runoff coefficient.
- Artificial Intelligence (AI) and Machine Learning (ML) can help analyse rainfall data and predict the quantity of rainwater that can be collected from a particular area.
- The proposed AI-Based Rainwater Harvesting Potential Predictor aims to estimate collectible rainwater and help households, farmers, and institutions plan suitable storage systems. This project promotes water conservation and efficient water resource management.

2. Literature Survey

12 research papers (2017–2026) were reviewed:

- **GIS and ML:** Choke Watershed (2026); Upper Tekeze Basin (2026) – ML-based harvesting zone identification.
- **GIS / AHP / ML:** Jaipur (2026); Bangladesh (2025) – suitable site identification.
- **Rainwater Estimation:** Shrigonda (2024); South Indian University (2021) – rooftop harvesting potential.
- **AI-Based Water Management:** Purulia (2024); Pune (2022) – water scarcity and groundwater recharge.
- **GIS and Rooftop Assessment:** Mouli et al. (2020); Pandya et al. (2018) – rooftop analysis and harvesting planning.
- **Sustainable Water Management:** Semi-Arid Tropics (2022); Singh and Turkiya (2017) – sustainable and rural water supply.

Detailed comparison is shown in Table 2.1 (next two slides).

Table 2.1: Literature Review (1/2)

No.	Paper (Author, Year)	Approach Type	Technologies Used	Key Objective
1	Choke Watershed Study (2026)	Ensemble ML	RF, GB, XGBoost	Identify suitable harvesting zones
2	Upper Tekeze Basin Study (2026)	Geo-hydrological ML	GIS, RF	Map potential harvesting zone
3	Jaipur Study (2026)	Integrated GIS-ML	GIS, AHP, ML	Identify suitable harvesting sites
4	Bangladesh Study (2025)	Comparative ML	RF, BRT, KNN, NB, AHP	Compare zone identification methods
5	Shrigonda Study (2024)	Potential Estimation	Rainfall, rooftop area	Estimate rooftop rainwater potential
6	Purulia Study (2024)	AI and GIS	ANN, RF, GIS	Assess harvesting suitability

Table 2.1: Literature Review (2/2)

No.	Paper (Author, Year)	Approach Type	Technologies Used	Key Objective
7	Mouli et al. (2020)	Rooftop Assessment	Satellite imagery, GIS	Estimate rooftop harvesting potential
8	South Indian University Study (2021)	Feasibility Analysis	Rainfall, roof area	Estimate annual collection potential
9	Vaddadi et al. (2022)	Groundwater Recharge	Rooftop harvesting analysis	Assess groundwater recharge potential
10	Semi-Arid Tropics Study (2022)	Sustainable Planning	Rainfall-runoff analysis	Identify harvesting interventions
11	Singh and Turkiya (2017)	Rural Water Supply	Rooftop harvesting assessment	Evaluate domestic water potential
12	Pandya et al. (2018)	GIS-Based Planning	Google Earth, QGIS	Plan rooftop harvesting structures

3. Information to seminar topic

- **Overview:** AI-based prediction of rainwater harvesting potential.
- **Rainwater Factors:** Rainfall, rooftop area, runoff coefficient, soil type, and slope.
- **Need:** Water conservation and reduction of water scarcity.
- **Working Principle:** Input Data → Pre-processing → ML Model → Prediction.
- **Architecture:** Data Collection, Pre-processing, Prediction Model, and Result Display.
- **Techniques:** Random Forest, Gradient Boosting, XGBoost, ANN, and GIS.
- **Evaluation Metrics:** Accuracy, precision, recall, and F1-score.
- **Challenges:** Limited data, rainfall variation, and prediction errors.

4. Advantages / Disadvantages

Advantages

- Helps conserve water and reduce water scarcity.
- Estimates rainwater collection potential.
- Supports better harvesting system planning.
- Saves time through automatic prediction.
- Helps identify suitable harvesting zones.
- Supports sustainable water management.

Disadvantages

- Requires accurate rainfall and rooftop data.
- Prediction may be inaccurate.
- Weather changes affect results.
- GIS data may not always be available.
- ML models require sufficient data.
- Results depend on the selected model and input data.

9

5. Conclusion

- AI-Based Rainwater Harvesting Potential Predictor is a useful approach for estimating rainwater collection potential using rainfall, rooftop area, and environmental data.
- Machine Learning techniques such as Random Forest, Gradient Boosting, and XGBoost can help improve prediction, while GIS supports the identification of suitable rainwater harvesting zones.
- This system can help households, colleges, farms, and urban areas plan rainwater harvesting more effectively. It supports water conservation, reduces water wastage, and helps address water scarcity.
- However, prediction accuracy depends on the quality of input data and changing weather conditions. Overall, this approach promotes sustainable water management and efficient use of rainwater resources.

10

6. Project Initiation

- **Project:** AI-Based Rainwater Harvesting Potential Predictor
- **Aim:** A system that uses rainfall, rooftop area, and environmental data to predict rainwater harvesting potential.

This section covers:

- 6.1 Description of Problem Statement
- 6.2 Scope and Objectives
- 6.3 Stakeholders
- 6.4 Seminar Discussion Plan
- 6.5 Platform / Equipment / Resources

11

6. 1 Description of problem statement

- Rainwater is often wasted due to a lack of proper collection and storage systems.
- It is difficult to estimate rainwater harvesting potential accurately.
- Traditional methods may require manual calculations and environmental data.
- Therefore, an AI-based system is needed to predict rainwater harvesting potential using rainfall, rooftop area, and other factors.
- This system can help users plan efficient rainwater collection and reduce water scarcity.

12

6. 2 Scope and objectives.

Scope

- Develop an AI-based system to predict rainwater harvesting potential.
- Estimate rainwater collection using rainfall and rooftop area.
- Consider environmental factors such as soil type and slope.
- Display the predicted rainwater collection potential.
- Support homes, colleges, farms, and urban areas.

Objectives

- To study existing rainwater harvesting methods.
- To collect rainfall and rooftop area data.
- To apply Machine Learning techniques for prediction.
- To identify suitable rainwater harvesting locations using GIS.
- To support water conservation and efficient rainwater management.

13

6. 3 Stake holder

- **Homeowners:** Estimate rainwater collection potential for their rooftops.
- **Farmers:** Plan rainwater harvesting for agricultural use.
- **Engineers and Planners:** Identify suitable harvesting locations.
- **Government Departments:** Support water conservation planning.
- **Researchers:** Study AI-based rainwater prediction methods.
- **Project Guide:** Guide the project and evaluate its progress.

14

6. 4 seminar exsicussion plan

- Introduction to AI-Based Rainwater Harvesting Potential Prediction.
- Need for AI in rainwater harvesting.
- Literature review of 12 research papers and comparison.
- Working principle and system architecture.
- Techniques, technologies, and evaluation metrics.
- Advantages, disadvantages, and challenges.
- Conclusion and project initiation.
- Question and answer session.

15

6.5 Platform/ Equipment/ Resources

Platform

- Windows or Linux operating system.
- Python programming language.
- Visual Studio Code for development.
- Google Colab for ML model training, if required.

Equipment

- Computer or laptop with a modern processor.
- Minimum 8 GB RAM recommended.
- Internet connection for datasets and libraries.
- Sufficient storage for data and model files.

16

Table 6.1: Resources

Resources	Name	Purpose
Operating System	Windows 11	Run development tools and software.
Programming Language	Python 3.11	Develop prediction models and application code.
Code Editor	Visual Studio Code	Write, debug, and test code.
ML Library	Scikit-learn	Build and evaluate ML models.
ML Algorithm	Random Forest, XGBoost	Predict rainwater harvesting potential.
Data Handling	Pandas, NumPy	Clean and process rainfall data.
Data Visualization	Matplotlib	Display data and prediction results.
GIS Tools	QGIS	Identify suitable harvesting zones.
Dataset	Rainfall and rooftop area data	Train and test prediction models.

17

Bibliography

Books

1. Rainwater Harvesting: In Urban Centers within the Hard Rock Terrain of the Deccan Basalts of India — Anil Lalwani, Springer, 2022.
2. Rainwater Harvesting—Building a Water Smart City — Aysha Akter, Springer, 2022.
3. Harvesting Rainwater from Buildings — Syed Azizul Haq, Springer, 2017.
4. Managed Groundwater Recharge and Rainwater Harvesting: Outlook from Developing Countries — Edited by Dipankar Saha, Karen G. Villholth, and Mohamed Shamrukh, Springer, 2024.

Websites

1. Google Scholar – Research papers on AI-based rainwater harvesting prediction and rainfall analysis.
2. ResearchGate – Research articles on rainwater harvesting, water conservation, and machine learning.
3. IEEE Xplore – Technical papers on Artificial Intelligence, Machine Learning, and predictive modelling.
4. ScienceDirect – Research articles on rainwater harvesting, GIS techniques, and machine learning-based prediction.

18

References

1. "Rainwater Harvesting Potential Zone Identification Using GIS – Machine Learning Approach and Comparison of Their Performance," 2026.
2. "Integrating Machine Learning and Geo-Hydrological Modeling to Identify Rainwater Harvesting Potential Zones in the Upper Tekeze Basin," 2026.
3. "Integrated Machine Learning and AHP-Based Framework for Spatial Delineation of Rainwater Harvesting Potential," 2026.
4. "Assessing the Performance of Machine Learning and AHP Models for Rainwater Harvesting Potential Zone Identification in Hilly Region, Bangladesh," 2025.
5. "Estimation of Roof Top Rain Water Harvesting Potential: A Proposed Solution to Water Scarcity in Shrigonda Town, Maharashtra, India," 2024.
6. "Addressing Water Scarcity Challenges Through Rainwater Harvesting: A Case Study on Purulia, India," 2024.
7. T. Mouli, K. Krupavathi and Ch. S. Ganeswara Rao, "Urban Rooftop Area Assessment for Estimation of Rooftop Rainwater-Harvesting Potential," 2020.
8. "Feasibility of Roof Top Rainwater Harvesting Potential – A Case Study of South Indian University," 2021.
9. N. Vaddadi, C. Vansarochana and V. Raghavan, "Estimation of Groundwater Recharge Potential Using Rooftop Rainwater Harvesting: Case Study from Pune Urban Area, India," 2022.
10. "Identifying Potential Zones for Rainwater Harvesting Interventions for Sustainable Intensification in the Semi-Arid Tropics," 2022.
11. Omvir Singh and Sushila Turkiya, "Assessing Potential for Rooftop Rainwater Harvesting: An Option for Sustainable Rural Domestic Water Supply in Arid Region of Haryana," 2017.
12. P. A. Pandya et al., "Planning of Rooftop Rainwater Harvesting Structure Based on GIS Approach," 2018.

19

Thank you

BIBLIOGRAPHY

- **Books**

1. Rainwater Harvesting: In Urban Centers within the Hard Rock Terrain of the Deccan Basalts of India — Anil Lalwani, Springer, 2022.
2. Rainwater Harvesting—Building a Water Smart City — Aysha Akter, Springer, 2022.
3. Harvesting Rainwater from Buildings — Syed Azizul Haq, Springer, 2017.
4. Managed Groundwater Recharge and Rainwater Harvesting: Outlook from Developing Countries — Edited by Dipankar Saha, Karen G. Villholth, and Mohamed Shamrukh, Springer, 2024.

- **Websites**

1. Google Scholar – Research papers related to AI-based rainwater harvesting prediction, rainfall analysis, and Machine Learning.
2. ResearchGate – Research articles on rainwater harvesting potential, water conservation, and machine learning models.
3. IEEE Xplore – Technical research papers related to Artificial Intelligence, Machine Learning, and predictive modelling.
4. ScienceDirect – Research articles on rainwater harvesting potential assessment, GIS techniques, and machine learning-based prediction.
5. Central Ground Water Board (CGWB) – Official information, guidelines, and publications related to rainwater harvesting and groundwater recharge.
6. India Meteorological Department (IMD) – Weather information and rainfall-related data useful for understanding rainfall patterns.

REFERENCES

1. “Rainwater Harvesting Potential Zone Identification Using GIS – Machine Learning Approach and Comparison of Their Performance,” 2026.
2. “Integrating Machine Learning and Geo-Hydrological Modeling to Identify Rainwater Harvesting Potential Zones in the Upper Tekeze Basin,” 2026.
3. “Integrated Machine Learning and AHP-Based Framework for Spatial Delineation of Rainwater Harvesting Potential,” 2026.
4. “Assessing the Performance of Machine Learning and AHP Models for Rainwater Harvesting Potential Zone Identification in Hilly Region, Bangladesh,” 2025.
5. “Estimation of Roof Top Rain Water Harvesting Potential: A Proposed Solution to Water Scarcity in Shrigonda Town, Maharashtra, India,” 2024.
6. “Addressing Water Scarcity Challenges Through Rainwater Harvesting: A Case Study on Purulia, India,” 2024.
7. T. Mouli, K. Krupavathi and Ch. S. Ganeswara Rao, “Urban Rooftop Area Assessment for Estimation of Rooftop Rainwater-Harvesting Potential,” 2020.
8. “Feasibility of Roof Top Rainwater Harvesting Potential – A Case Study of South Indian University,” 2021.
9. N. Vaddadi, C. Vansarochana and V. Raghavan, “Estimation of Groundwater Recharge Potential Using Rooftop Rainwater Harvesting: Case Study from Pune Urban Area, India,” 2022.
10. “Identifying Potential Zones for Rainwater Harvesting Interventions for Sustainable Intensification in the Semi-Arid Tropics,” 2022.
11. Omvir Singh and Sushila Turkiya, “Assessing Potential for Rooftop Rainwater Harvesting: An Option for Sustainable Rural Domestic Water Supply in Arid Region of Haryana,” 2017.
12. P. A. Pandya et al., “Planning of Rooftop Rainwater Harvesting Structure Based on GIS Approach,” 2018.

SEMINAR and PROJECT INITIATION Assessment / Evaluation

Formative Assessment

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960288	388588	Hemangi Shivaji Suryawanshi
2	24610960289	388589	Kalyani Naresh Thakare
3	24610960293	388593	Vidhi Shekhar Vibhandik
4	25610960008	388606	Shrawani Nutan Gautam

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388588	388589	388593	388606
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name: Mr. Nishad D. Patel

(Course Expert/s)